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Complete 29 Sprint × 7 Day Study Plan

Generated automatically from the TakeUforward Planly page.

Total sprints: 29

Sprint 1

Day 1

Task	Estimated Time
Linear Search	4 min
Largest Element	5 min

Task	Estimated Time
Second Largest Element	15 min
Java Basics	1h 36m

Day 2

Task	Estimated Time
Maximum Consecutive Ones	3 min
Left Rotate Array by One	6 min
Left Rotate Array by K Places	17 min
Move Zeros to End	13 min
Remove duplicates from sorted array	11 min

Day 3

Task	Estimated Time
Find missing number	18 min
Union of two sorted arrays	16 min
Intersection of two sorted arrays	13 min

Day 4

Task	Estimated Time
Majority Element-I	17 min
Leaders in an Array	11 min
Rearrange array elements by sign	9 min
Print the matrix in spiral manner	15 min

Day 5

Task	Estimated Time
Pascal's Triangle I	10 min
Pascal's Triangle II	9 min
Pascal's Triangle III	7 min
Rotate matrix by 90 degrees	17 min
Two Sum	17 min

Day 6

Task	Estimated Time
3 Sum	37 min

Day 7

Task	Estimated Time
4 Sum	27 min
Sort an array of 0's 1's and 2's	24 min
Kadane's Algorithm	19 min
Next Permutation	26 min
Majority Element-II	25 min
What is OOPS	16 min
Classes and Objects	15 min

Sprint 2

Day 1

Task	Estimated Time
Find the repeating and missing number	41 min
Count Inversions	23 min
Practice (Classes and Objects)	45 min

Day 2

Task	Estimated Time
Reverse Pairs	31 min
Maximum Product Subarray in an Array	18 min

Day 3

Task	Estimated Time
Merge two sorted arrays without extra space	32 min
Attributes and Methods	16 min

Day 4

Task	Estimated Time
Selection Sort	15 min
Bubble Sort	14 min
Insertion Sorting	11 min

Day 5

Task	Estimated Time
Practice (Attributes and Methods)	45 min

Day 6

Task	Estimated Time
Merge Sorting	47 min

Day 7

Task	Estimated Time
Quick Sorting	33 min
Basic Hashing	1h 34m
Longest Consecutive Sequence in an Array	22 min
Constructors	22 min

Sprint 3

Day 1

Task	Estimated Time
Longest subarray with sum K	40 min
Count subarrays with given sum	22 min
Practice (Constructors)	45 min

Day 2

Task	Estimated Time
Count subarrays with given xor K	23 min
Reverse a String II	9 min
Palindrome Check	7 min
Largest Odd Number in a String	11 min

Day 3

Task	Estimated Time
Longest Common Prefix	11 min
Isomorphic Strings	16 min
Rotate String	13 min
Valid Anagram	8 min
Encapsulation	8 min

Day 4

Task	Estimated Time
Sort Characters by Frequency	12 min
Practice (Encapsulation)	45 min

Day 5

Task	Estimated Time
Search X in sorted array	33 min
Lower Bound	16 min

Task	Estimated Time
Upper Bound	4 min
Search insert position	3 min

Day 6

Task	Estimated Time
Floor and Ceil in Sorted Array	6 min
First and last occurrence	22 min
Search in rotated sorted array-I	15 min
Search in rotated sorted array-II	12 min

Day 7

Task	Estimated Time
Find minimum in Rotated Sorted Array	15 min
Find out how many times the array is rotated	4 min
Single element in sorted array	22 min
Find square root of a number	17 min
Find Nth root of a number	20 min
Find the smallest divisor	15 min
Koko eating bananas	20 min
Access Modifiers	17 min
Practice (Access Modifiers)	45 min

Sprint 4

Day 1

Task	Estimated Time
Minimum days to make M bouquets	25 min
Aggressive Cows	26 min
Book Allocation Problem	27 min
Inheritance	36 min

Day 2

Task	Estimated Time
Find peak element	32 min

Day 3

Task	Estimated Time
Practice (Inheritance)	45 min

Day 4

Task	Estimated Time
Median of 2 sorted arrays	50 min
Kth element of 2 sorted arrays	10 min

Day 5

Task	Estimated Time
Minimize Max Distance to Gas Station	58 min

Day 6

Task	Estimated Time
Split array - largest sum	11 min
Find row with maximum 1's	10 min
Search in a 2D Matrix	15 min
Polymorphism	16 min

Day 7

Task	Estimated Time
Search in 2D matrix - II	15 min
Find Peak Element - II	20 min
Matrix Median	23 min
Recursion Theory	40 min
Recursion Concepts with Parameters	18 min
Practice (Polymorphism)	45 min

Sprint 5

Day 1

Task	Estimated Time
Pow(x,n)	23 min
Generate Parentheses	22 min
Power Set	20 min
Check if there exists a subsequence with sum K	23 min
Abstraction	29 min

Day 2

Task	Estimated Time
Count all subsequences with sum K	12 min
Practice (Abstraction)	45 min

Day 3

Task	Estimated Time
Combination Sum	16 min
Combination Sum II	15 min
Subsets I	7 min
Subsets II	21 min

Day 4

Task	Estimated Time
Combination Sum III	12 min
Letter Combinations of a Phone Number	12 min
Palindrome partitioning	18 min

Day 5

Task	Estimated Time
Word Search	19 min
N Queen	24 min

Day 6

Task	Estimated Time
Rat in a Maze	19 min
Interfaces	20 min

Day 7

Task	Estimated Time
M Coloring Problem	22 min
Sudoku Solver	31 min
Introduction to Singly LinkedList	37 min
Traversal in Linked List	3 min
Deletion in Linked List	32 min
Practice (Interfaces)	45 min

Sprint 6

Day 1

Task	Estimated Time
Insertion in Linked List	24 min
Deletion of the head of LL	6 min
Deletion of the tail of Linked List	10 min
Deletion of the Kth element of Linked List	15 min
Delete the element with value X	2 min
Insertion at the head of Linked List	3 min
Insertion at the tail of Linked List	4 min
Static Keyword	20 min

Day 2

Task	Estimated Time
Insertion at the Kth position of Linked List	13 min
Practice (Static Keyword)	45 min

Day 3

Task	Estimated Time
Insertion before the value X in Linked List	4 min
Introduction to Doubly LL	4 min
Deletion in Doubly LL	30 min
Insertion in DLL	15 min

Day 4

Task	Estimated Time
Convert Array to Doubly Linked List	10 min
Delete Tail of Doubly Linked List	5 min
Delete Kth Element of Doubly Linked List	14 min
Removing given node in Doubly Linked List	5 min
Insert node before head in Doubly Linked List	3 min

Task	Estimated Time
Insert node before tail in Doubly Linked List	5 min
Insert node before (kth node) in Doubly Linked List	5 min
Insert before given node in Doubly Linked List	2 min

Day 5

Task	Estimated Time
Add two numbers in Linked List	14 min
Segregate odd and even nodes in Linked List	24 min

Day 6

Task	Estimated Time
Sort a Linked List of 0's 1's and 2's	22 min
Inner classes	23 min

Day 7

Task	Estimated Time
Remove Nth node from the back of the LL	16 min
Reverse a LL	32 min
Add one to a number represented by LL	24 min
Find Middle of Linked List	14 min
Delete the middle node in LL	16 min
Check if LL is palindrome or not	20 min
Practice (Inner classes)	45 min

Sprint 7

Day 1

Task	Estimated Time
Find the intersection point of Y LL	32 min
Detect a loop in LL	20 min
Find the starting point in LL	21 min
Association, Aggregation, and Composition	23 min

Day 2

Task	Estimated Time
Length of loop in LL	14 min
Practice (Composition)	45 min

Day 3

Task	Estimated Time
Reverse LL in group of given size K	24 min
Rotate a LL	12 min
Merge two Sorted Lists	18 min

Day 4

Task	Estimated Time
Flattening of LL	33 min
Sort LL	22 min

Day 5

Task	Estimated Time
Clone a LL with random and next pointer	33 min
Delete all occurrences of a key in DLL	11 min
Object Cloning	15 min

Day 6

Task	Estimated Time
Remove duplicates from sorted DLL	12 min
Practice (Object Cloning)	45 min

Day 7

Task	Estimated Time
Introduction to Bits and Tricks	1h 20m
Minimum Bit Flips to Convert Number	7 min
Single Number - I	7 min
Single Number - II	30 min
Single Number - III	24 min

Sprint 8

Day 1

Task	Estimated Time
Divide two numbers without multiplication and division	19 min
Power Set Bit Manipulation	12 min
XOR of numbers in a given range	5 min
Theory	36 min
Maximum Points You Can Obtain from Cards	11 min
Exception Handling	34 min

Day 2

Task	Estimated Time
Longest Substring Without Repeating Characters	22 min
Generics	29 min

Day 3

Task	Estimated Time
Max Consecutive Ones III	29 min
Fruit Into Baskets	29 min

Day 4

Task	Estimated Time
Longest Substring With At Most K Distinct Characters	21 min
Longest Repeating Character Replacement	25 min

Day 5

Task	Estimated Time
Minimum Window Substring	26 min
File Handling	29 min

Day 6

Task	Estimated Time
Number of Substrings Containing All Three Characters	19 min
Binary Subarrays With Sum	20 min
Count number of Nice subarrays	4 min

Day 7

Task	Estimated Time
Implementation using different DS	1h 4m
Implement Stack using Arrays	8 min
Implement Queue using Arrays	12 min
Implement Stack using Queue	6 min
Implement Queue using Stack	14 min
Implement stack using Linkedlist	7 min
Implement queue using Linkedlist	7 min
Balanced Paranthesis	12 min
Design Principles	23 min
Object Lifecycle	18 min

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Day 1

Task	Estimated Time
Next Greater Element	18 min
Next Greater Element - 2	15 min
Asteroid Collision	17 min
Sum of Subarray Minimums	23 min
Sum of Subarray Ranges	10 min
Why Networks exists	9 min
Network Types	9 min
Internet and Backbone Networks	14 min

Day 2

Task	Estimated Time
Remove K Digits	15 min
Implement Min Stack	21 min
Clients, Servers and Peers	11 min

Day 3

Task	Estimated Time
Sliding Window Maximum	20 min
How data moves in packets	18 min

Day 4

Task	Estimated Time
Trapping Rainwater	29 min
Largest rectangle in a histogram	31 min

Day 5

Task	Estimated Time
Maximum Rectangles	12 min

Task	Estimated Time
Why do we need layered Architecture	9 min

Day 6

Task	Estimated Time
OSI Model	1h 4m

Day 7

Task	Estimated Time
Stock span problem	19 min
Celebrity Problem	16 min
LRU Cache	25 min
LFU Cache	45 min
Assign Cookies	10 min
Lemonade Change	9 min
Jump Game - I	10 min
Shortest Job First	7 min
Job sequencing Problem	16 min
N meetings in one room	13 min
Non-overlapping Intervals	8 min

Sprint 10

Day 1

Task	Estimated Time
Insert Interval	13 min
Minimum number of platforms required for a railway	18 min
Valid Paranthesis Checker	25 min
Candy	30 min
How Data Travels Up the OSI Model	5 min
TCP/IP Model	10 min
OSI vs TCP/IP	2 min
Encapsulation and Decapsulation	6 min
Physical Layer	11 min

Day 2

Task	Estimated Time
Introduction	32 min
Transmission Media	8 min

Day 3

Task	Estimated Time
Inorder Traversal	37 min
Twisted Pair Cable	11 min

Day 4

Task	Estimated Time
Preorder Traversal	18 min
Postorder Traversal	14 min
Level Order Traversal	8 min
Fiber Optic Cable	8 min
Coaxial Cable	6 min

Day 5

Task	Estimated Time
Pre, Post, Inorder in one traversal	10 min
Maximum Depth in BT	7 min
Check if two trees are identical or not	4 min
Full duplex vs Half duplex	4 min
Network Device	33 min

Day 6

Task	Estimated Time
Check for balanced binary tree	12 min
Diameter of Binary Tree	13 min
Maximum path sum	17 min
Check for symmetrical BTs	8 min
Zig Zag or Spiral Traversal	8 min

Day 7

Task	Estimated Time
Boundary Traversal	9 min
Vertical Order Traversal	18 min
Top View of BT	9 min
Bottom view of BT	12 min
Right/Left View of BT	12 min
Print root to leaf path in BT	10 min
LCA in BT	13 min
Maximum Width of BT	21 min
Print all nodes at a distance of K in BT	16 min
Minimum time taken to burn the BT from a given Node	17 min
Mac address	12 min
Error Detection	17 min
Medium Access and CSMA	13 min

Sprint 11

Day 1

Task	Estimated Time
Count total nodes in a complete BT	15 min
Requirements needed to construct a unique BT	7 min
Construct a BT from Preorder and Inorder	18 min
Construct a BT from Postorder and Inorder	19 min
Serialize and De-serialize BT	16 min
Morris Inorder Traversal	19 min
Address Resolution Protocol	17 min
Network Access Control	8 min

Day 2

Task	Estimated Time
Morris Preorder Traversal	4 min
Introduction to BST	7 min
Search in BST	5 min
Floor and Ceil in a BST	8 min
Why do we need STP?	30 min

Day 3

Task	Estimated Time
Insert a given node in BST	7 min
Delete a node in BST	15 min
Kth Smallest and Largest element in BST	8 min
Check if a tree is a BST or not	9 min
LCA in BST	8 min

Day 4

Task	Estimated Time
Construct a BST from a preorder traversal	16 min

Task	Estimated Time
Inorder successor and predecessor in BST	10 min
Spanning Tree Protocol (STP)	30 min

Day 5

Task	Estimated Time
BST iterator	14 min
Two sum in BST	14 min
Correct BST with two nodes swapped	16 min

Day 6

Task	Estimated Time
Largest BST in Binary Tree	17 min
Heaps (Theory Video)	27 min
Network Topologies and VLANs Introduction	4 min

Day 7

Task	Estimated Time
Heapify Algorithm	23 min
Build heap from a given Array	15 min
Implement Min Heap	12 min
Implement Max Heap	20 min
Check if an array represents a min heap	7 min
Convert Min Heap to Max Heap	4 min
Heap Sort	30 min
Physical Topologies	23 min
Hybrid Topology	4 min
Logical Topologies	8 min
Ethernet Switching	13 min

Sprint 12

Day 1

Task	Estimated Time
K-th Largest element in an array	44 min
Kth largest element in a stream of running integers	16 min
Virtual LANs	19 min
Switching vs Routing	3 min

Day 2

Task	Estimated Time
Introduction to Graph	41 min
Introduction to Network Layer	14 min

Day 3

Task	Estimated Time
Traversal Techniques	39 min

Day 4

Task	Estimated Time
Connected Components	7 min
Number of provinces	15 min
IPv4 Introduction	27 min

Day 5

Task	Estimated Time
Number of islands	24 min
Flood fill algorithm	20 min

Day 6

Task	Estimated Time
Number of enclaves	15 min

Task	Estimated Time
IPv4 Datagram	28 min

Day 7

Task	Estimated Time
Rotten Oranges	22 min
Distance of nearest cell having one	20 min
Surrounded Regions	23 min
Number of distinct islands	18 min
Detect a cycle in an undirected graph	38 min
Bipartite graph	32 min
IP Addressing Modes	9 min
Private vs Public IP	10 min
Classful Addressing	7 min

Sprint 13

Day 1

Task	Estimated Time
Topological sort or Kahn's algorithm	26 min
Detect a cycle in a directed graph	23 min
Find eventual safe states	39 min
Subnetting	11 min
Common Interview Questions	16 min

Day 2

Task	Estimated Time
Course Schedule I	11 min
Course Schedule II	11 min
Alien Dictionary	20 min
CIDR - Classless Addressing	16 min

Day 3

Task	Estimated Time
Shortest path in DAG	26 min
IPv6 Introduction	23 min

Day 4

Task	Estimated Time
Shortest path in undirected graph with unit weights	16 min
Word ladder I	28 min
IPv6 Datagram	14 min

Day 5

Task	Estimated Time
Word ladder II	48 min
IPv6 over IPv4	5 min

Day 6

Task	Estimated Time
Dijkstra's algorithm	47 min
Router vs Forwarding	13 min

Day 7

Task	Estimated Time
Print Shortest Path	19 min
Shortest Distance in a Binary Maze	21 min
Path with minimum effort	24 min
Cheapest flight within K stops	23 min
Minimum multiplications to reach end	19 min
Number of ways to arrive at destination	23 min
Routing Tables Basics	11 min
Static Routing	5 min
Dynamic Routing	10 min
Distance Vector Routing and RIP	7 min
Routing Loop and Prevention	18 min

Sprint 14

Day 1

Task	Estimated Time
Bellman ford algorithm	27 min
Floyd warshall algorithm	30 min
Find the city with the smallest number of neighbors	12 min
MST theory	7 min
Link State Routing and OSPF	12 min
Hybrid Routing and EIGRP	3 min
Border Gateway Protocol	17 min

Day 2

Task	Estimated Time
Disjoint Set	41 min
Transport Layer Introduction	14 min

Day 3

Task	Estimated Time
Find the MST weight	31 min
Port and Sockets	17 min

Day 4

Task	Estimated Time
Number of operations to make network connected	14 min
Accounts merge	21 min
Number of islands II	25 min

Day 5

Task	Estimated Time
Making a large island	26 min
UDP	10 min

Day 6

Task	Estimated Time
Most stones removed with same row or column	23 min
TCP-I: Connection Setup and Numbering	28 min

Day 7

Task	Estimated Time
Kosaraju's algorithm	22 min
Bridges in graph	23 min
Articulation point in graph	21 min
Introduction to DP	30 min
Climbing stairs	13 min
Frog Jump	37 min
TCP-II: Reliability, Flow Control and Sliding Window	15 min
TCP-III: Congestion Control and Connection Closing	17 min

Sprint 15

Day 1

Task	Estimated Time
Frog jump with K distances	16 min
Maximum sum of non adjacent elements	31 min
House robber	9 min
When to choose TCP/UDP	5 min
Application Layer Introduction	4 min
HTTP-Request Response Model	7 min
HTTP methods and status codes	17 min

Day 2

Task	Estimated Time
Ninja's training	51 min

Day 3

Task	Estimated Time
HTTP Versions	31 min

Day 4

Task	Estimated Time
Grid unique paths	44 min
Unique paths II	12 min

Day 5

Task	Estimated Time
Minimum Falling Path Sum	41 min
HTTP/3 and QUIC	14 min

Day 6

Task	Estimated Time
Triangle	33 min
HTTPS and TLS	15 min

Day 7

Task	Estimated Time
Cherry pickup II	42 min
Best time to buy and sell stock	7 min
Best time to buy and sell stock II	35 min
Best time to buy and sell stock III	25 min
Best time to buy and sell stock IV	12 min
Best time to buy and sell stock with transaction fees	6 min
Cookies, Sessions, and tokens	22 min
Rest API's	9 min

Sprint 16

Day 1

Task	Estimated Time
Subset sum equals to target	35 min
Partition equal subset sum	7 min
Partition a set into two subsets with minimum absolute sum difference	29 min
WebSockets	8 min
File Transfer Protocol	4 min
Email Protocols	7 min
NAT and Why do we need it?	10 min

Day 2

Task	Estimated Time
Count subsets with sum K	35 min
NAT Packet Flow	9 min

Day 3

Task	Estimated Time
Count partitions with given difference	17 min
Types of NAT	7 min

Day 4

Task	Estimated Time
0 and 1 Knapsack	40 min
SNAT, DNAT and Port Forwarding	7 min
NAT Limitations	13 min

Day 5

Task	Estimated Time
Minimum coins	34 min
Target sum	8 min

Task	Estimated Time
DNS	17 min

Day 6

Task	Estimated Time
Coin change II	22 min
Unbounded knapsack	22 min
DNS Record Types	6 min

Day 7

Task	Estimated Time
Rod cutting problem	22 min
Longest Increasing Subsequence	40 min
Print Longest Increasing Subsequence	26 min
Largest Divisible Subset	14 min
Longest String Chain	16 min
Longest Bitonic Subsequence	13 min
DNS Security	6 min
DNSSEC	6 min
Dynamic Host Configuration Protocol (DHCP)	10 min
What is Switching and Why?	6 min
Fundamental Switching Techniques	4 min
Circuit Switching	13 min

Sprint 17

Day 1

Task	Estimated Time
Number of Longest Increasing Subsequences	20 min
Longest common subsequence	46 min
Longest common substring	14 min
Longest palindromic subsequence	9 min
Packet Switching	22 min
Datagram Packet Switching	8 min

Day 2

Task	Estimated Time
Minimum insertions to make string palindrome	12 min
Minimum insertions or deletions to convert string A to B	7 min
Shortest common supersequence	26 min
Virtual Circuit Packet Switching	11 min

Day 3

Task	Estimated Time
Distinct subsequences	40 min
MPLS	6 min
Circuit Switching vs. Virtual Packet Switching	4 min

Day 4

Task	Estimated Time
Edit distance	37 min
Message Switching	12 min
Why Packet Switching became backbone of Internet	3 min
Comparisons between all 3	2 min
FAQs - Switching	4 min

Day 5

Task	Estimated Time
Wildcard matching	43 min
Why Network Security?	6 min
CIA Triad	11 min

Day 6

Task	Estimated Time
Matrix chain multiplication	1h 1m

Day 7

Task	Estimated Time
Burst balloons	33 min
Palindrome partitioning II	22 min
Trie Implementation and Operations	30 min
Trie Implementation and Advanced Operations	22 min
AAA Framework	10 min
Common Network Security Threats	19 min
Firewalls and Types	19 min
Firewall Type Comparison - L3, L4, and L7	3 min
IDPS	6 min

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Day 1

Task	Estimated Time
Longest Word with All Prefixes	25 min
Number of distinct substrings in a string	17 min
Maximum XOR of two numbers in an array	34 min
VPNs	16 min
Zero Trust Architecture	6 min
Introduction of Cryptography	4 min

Day 2

Task	Estimated Time
Maximum Xor with an element from an array	23 min
Security Goals of Cryptography	2 min
Basic Cryptography Terms	3 min
Symmetric Encryption	5 min

Day 3

Task	Estimated Time
Reverse every word in a string	31 min
Asymmetric Cryptography	14 min

Day 4

Task	Estimated Time
Minimum number of bracket reversals to make an expression balanced	20 min
Count and say	19 min
Diffie Hellman Key Exchange Method	11 min

Day 5

Task	Estimated Time
Rabin Karp Algorithm	34 min

Task	Estimated Time
Hybrid Encryption	4 min
Hash Functions	16 min

Day 6

Task	Estimated Time
Z function	35 min

Day 7

Task	Estimated Time
KMP Algorithm or LPS array	29 min
Shortest Palindrome	7 min
Longest happy prefix	3 min
Print all primes till N	18 min
Prime factorisation of a Number	15 min
Count primes in range L to R	16 min
MAC and HMAC	11 min
Digital Signatures	7 min
Certificate Authority and Chain of Trust	7 min
Public Key Infrastructure	3 min
Certificate Pinning	4 min
TLS and SSL	6 min
TLS Handshake and Forward Secrecy	6 min
Wireless Networking Introduction	6 min
Basics of Wireless Networking	17 min
Wi-Fi Joining Flow	11 min

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Day 1

Task	Estimated Time
WPA and WPA/WPA-2/WPA-3 Security Handshake	19 min
IEEE 802.11 and Wi-Fi Generations	8 min
Why Wi-Fi speed drops in Real Life	8 min
How Wi-Fi avoids Collisions	9 min
Wi-Fi Frame Types	4 min
Wi-Fi Security	9 min
Packet Flow: Opening a website over Wi-Fi	5 min
Introduction to Network Performance	11 min
Introduction to Low Level Design	21 min
Software Design Principles	23 min

Day 2

Task	Estimated Time
Metrics Deep Dive	12 min
Data Transfer and Bottlenecks, BDP	10 min
Latency Sources, Delay, and TTFB	15 min
Congestion, Bufferbloat, and Congestion Collapse Loop	9 min
QoS, Traffic Shaping, and Policing	8 min
Rate Limiting	6 min

Day 3

Task	Estimated Time
Fixed Window Counter - Rate Limiting Algo	5 min
Sliding Window Log - Rate Limiting Algo	4 min
Sliding Window Counter - Rate Limiting Algo	6 min
Token Bucket - Rate Limiting Algo	3 min
Leaky Bucket - Rate Limiting Algo	3 min

Task	Estimated Time
Concurrency Limiter - Rate Limiting Algo	2 min
Load Balancing Fundamentals	10 min
L4 vs L7 Load Balancers	4 min
Single Responsibility Principle (SRP)	16 min

Day 4

Task	Estimated Time
Load Balancing Algorithms	30 min
Open Closed Principle (OCP)	17 min

Day 5

Task	Estimated Time
CDN Caching and Lifestyle	30 min
What happens when you type google.com	30 min

Day 6

Task	Estimated Time
What happens when DNS fails?	30 min
Liskov Substitution Principle (LSP)	18 min
Interface Segregation Principle (ISP)	10 min

Day 7

Task	Estimated Time
What happens when you connect to public Wi-Fi?	30 min
How does a CDN serve an image/video?	30 min
Dependency Inversion Principle (DIP)	14 min
Unified Modeling Language (UML)	10 min
Class UML diagrams	38 min
Introduction to Design Patterns	10 min
Singleton Design Pattern	25 min
Factory Method	16 min

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Day 1

Task	Estimated Time
Builder Pattern	24 min
Abstract Factory	26 min
Prototype Pattern	20 min
Introduction to SQL	42 min

Day 2

Task	Estimated Time
Adapter Pattern	17 min
Decorator Pattern	26 min
Facade Pattern	17 min

Day 3

Task	Estimated Time
Composite Pattern	21 min
Proxy Pattern	17 min
Why SQL Exists	12 min

Day 4

Task	Estimated Time
Bridge Pattern	18 min
Flyweight Pattern	19 min
How Databases Work	19 min

Day 5

Task	Estimated Time
Iterator Pattern	25 min
Observer Pattern	18 min
Database Systems	10 min

Day 6

Task	Estimated Time
Strategy Pattern	12 min
Command Pattern	28 min
Installation and Tools	16 min

Day 7

Task	Estimated Time
Template Method	17 min
State Pattern	21 min
Chain of Responsibility	17 min
Visitor Pattern	23 min
Mediator Pattern	14 min
Memento Pattern	17 min
File Systems vs Database Management Systems	15 min
Data Abstraction, Schemas, Instances, and Data Independence	15 min
DBMS Components and End-to-End Query Flow	15 min

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Day 1

Task	Estimated Time
Multithreading and Concurrency	40 min
Creating and Managing Threads	35 min
Database Users, Roles, and Deployment Architectures	15 min
Data Models and Database Families	15 min

Day 2

Task	Estimated Time
Thread Pools and Executors	32 min

Day 3

Task	Estimated Time
Database & Table Basics	1h 31m

Day 4

Task	Estimated Time
Thread Safety and Synchronization	29 min
Locks and Synchronization Mechanism	31 min

Day 5

Task	Estimated Time
Deadlock and Prevention Techniques	26 min
Producer Consumer Problem	17 min

Day 6

Task	Estimated Time
Dependency Injection	19 min
Exception Handling (LLD)	23 min

Day 7

Task	Estimated Time
Building Resilient Systems	19 min
All About API's	55 min
Database Design and Integration	24 min
How to approach a LLD Interview	36 min
SQL Basics and Commands	4 min
Working with Databases in SQL	9 min

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Day 1

Task	Estimated Time
Parking Lot (Design)	38 min
Parking Lot (Code)	29 min
SQL Data Types	6 min
Creating and Managing Tables in SQL	14 min
Primary Key	18 min

Day 2

Task	Estimated Time
Logging Framework (Design)	38 min
Logging Framework (Code)	22 min

Day 3

Task	Estimated Time
Foreign Key	18 min

Day 4

Task	Estimated Time
Traffic Signal System (Design)	1h

Day 5

Task	Estimated Time
Traffic Signal System (Code)	24 min
Constraints in SQL	16 min

Day 6

Task	Estimated Time
Vending Machine Design	47 min
NULL vs 0 vs Empty String	2 min

Task	Estimated Time
DDL vs DML	3 min
Query Lifecycle	4 min

Day 7

Task	Estimated Time
Vending Machine Code	27 min
Task Management System Design	51 min
Task Management System Code	22 min
Indexing in SQL	14 min
Entities, Entity Sets, and Attributes	15 min
Relationships and Structural Constraints	15 min

Sprint 23

Day 1

Task	Estimated Time
PubSub System Design	42 min
PubSub System Code	25 min
Weak Entities and Enhanced ER Modeling	15 min
ER/EER-to-Relational Mapping	15 min

Day 2

Task	Estimated Time
ATM Machine Design	48 min

Day 3

Task	Estimated Time
Database Design Fundamentals	1h 50m

Day 4

Task	Estimated Time
ATM Machine Code	24 min

Day 5

Task	Estimated Time
Hotel Management System Design	51 min

Day 6

Task	Estimated Time
Hotel Management System Code	19 min

Day 7

Task	Estimated Time
Elevator System Design	44 min

Task	Estimated Time
Elevator System Code	26 min
Digital Wallet Design	27 min
Types of Locking Mechanism	24 min
Digital Wallet Code	19 min

Sprint 24

Day 1

Task	Estimated Time
Ride Booking App Design	45 min
Ride Booking App Code	24 min
Importance, Raw Data and Pain	12 min
Core Concepts (Overview)	5 min
Mental Model (First Principle)	11 min

Day 2

Task	Estimated Time
Music Streaming Platform Design	42 min

Day 3

Task	Estimated Time
Streaming Protocols	14 min
Schema Design	29 min

Day 4

Task	Estimated Time
Music Streaming Platform Code	27 min
Entity and Attributes	3 min
Relationships, Cardinality and Optionality	15 min
Everything About Keys	10 min

Day 5

Task	Estimated Time
Normalisation and their Forms	24 min
Formal Relational Model and Integrity Constraints	15 min
Fundamental Relational Algebra	15 min

Day 6

Task	Estimated Time
Joins and Extended Relational Algebra	15 min
Tuple and Domain Relational Calculus	15 min
Views and Materialized Views	15 min
Triggers, Stored Routines, and Database-Side Logic	15 min

Day 7

Task	Estimated Time
Redundancy and Update Anomalies	15 min
Functional Dependencies and Keys	15 min
Armstrong's Axioms, Attribute Closure, and Minimal Cover	15 min
First, Second, Third, and Boyce–Codd Normal Forms	15 min
Decomposition Quality and Normalization Algorithms	15 min
Multivalued Dependencies, 4NF, Join Dependencies, and 5NF	15 min
Query Fundamentals	27 min
SELECT AND FROM	1 min
WHERE	2 min
Comparison Operators	2 min
Logical Operators	6 min
Arithmetic Operators	4 min
ORDER BY and LIMIT	2 min
DISTINCT and AS (Aliases)	2 min
Big Countries	15 min
Profitable Customers in 2021	15 min

Sprint 25

Day 1

Task	Estimated Time
Odd Non-Boring Movies	15 min
Filtering Essentials	24 min
IS NULL vs. IS NOT NULL, IN, and NOT IN	7 min
BETWEEN and NOT BETWEEN	6 min
LIKE and NOT LIKE	7 min
Filter Records Excluding a Specific Pattern	15 min
Find Records Excluding a Given Set of Values	15 min
Find Salaries Outside the Expected Range	15 min
Non-Referred Customers	15 min

Day 2

Task	Estimated Time
Data Summarization	39 min
Fundamentals of GROUP BY	11 min
Basic Aggregate Functions (MIN, MAX, SUM, AVG)	6 min

Day 3

Task	Estimated Time
COUNT Functions	14 min
HAVING Clause (Basics, HAVING with COUNT)	3 min
First Login Analysis	15 min
Employee Work Time Summary	15 min

Day 4

Task	Estimated Time
Unique Subjects per Teacher	15 min
User Follower Count	15 min
CRM Automotive Sales Analysis	15 min

Task	Estimated Time
Highest Order Placing Customer	15 min

Day 5

Task	Estimated Time
Frequent Actor-Director Duos	15 min
Large Classes	15 min
Email Duplicates	15 min

Day 6

Task	Estimated Time
Merging Query Results	17 min
UNION	8 min
UNION ALL	1 min
Difference Between UNION and UNION ALL	1 min
Intersection	6 min
Combine Active and Archived Users	15 min

Day 7

Task	Estimated Time
Merge Recent Orders from Multiple Sources	15 min
Combine Sales Records Without Deduplication	15 min
Reshape Products Data	15 min
Joins Deep Dive	1h 13m
RIGHT JOIN	23 min
ON vs. WHERE	15 min
LEFT JOIN	6 min
INNER JOIN	8 min
FULL OUTER JOIN	10 min

Sprint 26

Day 1

Task	Estimated Time
CROSS JOIN	2 min
IMPLICIT JOIN	4 min
SELF JOIN	10 min
NATURAL JOIN	5 min
Employees and Their Departments	15 min
Customer Orders Overview	15 min
Employees With Confirmed Salary Records	15 min
Generate All Possible User-Category Pairs	15 min
Employees With or Without Salary Records	15 min
Match Employees With Their Salaries	15 min

Day 2

Task	Estimated Time
Employees Earning More Than Their Manager	15 min
Students Enrolled in Courses	15 min
Sales Analysis	15 min
Minimum Distance Between Points	15 min

Day 3

Task	Estimated Time
Suspended Accounts	15 min
Find Team Size for Each Employee	15 min
Average Experience by Project	15 min
Warehouse Stock Manager	15 min

Day 4

Task	Estimated Time
Table Join Operation	15 min

Task	Estimated Time
Inactive Customers	15 min
Students Enrolled in Non-Existent Departments	15 min
Low Bonus Employees	15 min

Day 5

Task	Estimated Time
Available Seat Streaks	15 min
Visitors Without Transactions	15 min
A & B Buyers Without C	15 min
Product Selling Price Report	15 min

Day 6

Task	Estimated Time
Updated Bank Balances	15 min
Most Frequent Travellers	15 min
Suggested Pages	15 min

Day 7

Task	Estimated Time
Everything about Subqueries	1h 2m
Introduction to Subqueries (IN)	35 min
EXISTS and NOT EXISTS	15 min
Correlated Subqueries	11 min
Contest Participation Rate	15 min
High-Report Managers	15 min
All-Product Buyers	15 min

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Day 1

Task	Estimated Time
Highest Non-Repeating Number	15 min
Find the First Device Logged In by Each Player	15 min
Salespersons Without RED Orders	15 min
Employees with the Highest Salary in Each Department	15 min
Find the Most Recent Order for Each Product	15 min
Find Transactions with Maximum Amount Per Day	15 min
Incomplete Employee Records	15 min
Find Quiet Students in All Exams	15 min

Day 2

Task	Estimated Time
Top Grade per Student	15 min
Swap Consecutive Seats	15 min
Safe Investment Countries	15 min
Tennis Grand Slam Winners	15 min

Day 3

Task	Estimated Time
Football Team Scores	15 min
Order Count per Customer	15 min
Boolean Expression Evaluator	15 min
Immediate First Orders Percentage	15 min

Day 4

Task	Estimated Time
Find All Employees Reporting to the Head of the Company	15 min
Orphan Employees	15 min

Day 5

Task	Estimated Time
Editing Data and Tables	38 min
INSERT	6 min
UPSERT	7 min
UPDATE	5 min
DELETE	3 min

Day 6

Task	Estimated Time
ALTER	12 min
TRUNCATE	1 min
DELETE vs. TRUNCATE vs. DROP	2 min
System Settings	15 min
Employee Salary	15 min
Storage Hierarchy, Pages, Records, and Buffers	15 min

Day 7

Task	Estimated Time
File Organization	15 min
Index Classification and Design	15 min
B-Trees and B+ Trees	15 min
Static and Dynamic Hashing	15 min
Storage, Keys and Query Performance	1h 53m

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Day 1

Task	Estimated Time
Where Rows Actually Live	56 min
How a B+ Tree Is Built with Data	29 min
Why the Wrong Primary Key Can Quietly Destroy You	16 min
Index Strategy at Scale	13 min

Day 2

Task	Estimated Time
From SQL to an Executable Plan	15 min
Algorithms for Selection, Projection, Sorting, and Aggregation	15 min
Join Processing Algorithms	15 min
Cost-Based Query Optimization	15 min

Day 3

Task	Estimated Time
Performance & Debugging	1h 21m

Day 4

Task	Estimated Time
Raw Data Setup and Stored Procedures	37 min
Debugging Queries with EXPLAIN	22 min

Day 5

Task	Estimated Time
Query Performance	9 min
Debugging Correctness	12 min

Day 6

Task	Estimated Time
Permissions and Transactions Part-1	41 min

Day 7

Task	Estimated Time
Privileges and Roles	20 min
GRANTS	13 min
GRANT ALL and WITH GRANT OPTION	3 min
ALTER USER	1 min
REVOKE	1 min
Permissions and Transactions Part-2	1h 1m
Connection, Connection Pool and Commit	22 min
Rollback	1 min
Savepoint	5 min
Internal Working of Transactions, Timeout and Deadlock	33 min
Transaction Concepts, ACID, and States	15 min

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Day 1

Task	Estimated Time
Schedules and Concurrency Anomalies	15 min
Conflict and View Serializability	15 min
Recoverable, Cascadeless, and Strict Schedules	15 min
Lock-Based Protocols and Two-Phase Locking	15 min
Deadlock Handling	15 min
Timestamp Ordering, Optimistic Control, and MVCC	15 min
SQL Isolation Levels	15 min
Failures, Storage, and Recovery Goals	15 min

Day 2

Task	Estimated Time
Log-Based Recovery and Write-Ahead Logging	15 min
Checkpoints and ARIES Intuition	15 min
Shadow Paging, Backup, and Point-in-Time Recovery	15 min
Declarative Integrity Beyond Keys	15 min

Day 3

Task	Estimated Time
Authorization, Views, and Auditing	15 min
Database Security Threats and Protection	15 min
Metadata, Statistics, Maintenance, and Observability	15 min
Distributed Database Foundations	15 min

Day 4

Task	Estimated Time
Partitioning, Sharding, and Distributed IDs	15 min
Replication, Consistency, and CAP/PACELC	15 min
Distributed Transactions and Coordination	15 min

Task	Estimated Time
NoSQL Data Models and SQL-vs-NoSQL Decisions	15 min

Day 5

Task	Estimated Time
OLTP, OLAP, Warehousing, and Columnar Storage	15 min
Schema-Design Capstone	15 min
Storage and Query-Plan Lab	15 min
Concurrency and Recovery Lab	15 min

Day 6

Task	Estimated Time
GATE and Semester Problem Track	15 min
Interview Depth by Candidate Level	15 min
Final Coverage and Source Alignment	15 min